

ISABELLE GOLDSTEIN, PH.D

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EDUCATION

Brown University

Doctor of Philosophy, Physics
Master of Science, Physics

Providence, RI USA

2018–2023

2020

– Advisor: Savvas Koushiappas, PhD

Carnegie Mellon University

Bachelor of Science, Physics

Pittsburgh, PA USA

2014–2018

PUBLICATIONS

1. “Scaling Relations for Dark Matter Halos Hosting Ultra-Faint Dwarf Galaxies”
Levi K. C. Fisher, **Isabelle S. Goldstein**, Jason Kumar, and Louis E. Strigari
arXiv:2501.11138
2. “Kinematics of the Sagittarius Dwarf Spheroidal core: A 5D Analysis for a 6D Methodology with Gaia DR3”
Isabelle S. Goldstein and Louis E. Strigari
Mon Not R Astron Soc (2025) 1016-1027
3. “Viability of ultralight bosonic dark matter in dwarf galaxies”
Isabelle S. Goldstein, Savvas M. Koushiappas, and Matthew G. Walker
Phys. Rev. D 106, 063010 (2022)
4. “Could the $2.6 M_{\odot}$ object in GW190814 be a primordial black hole?”
Kyriakos Vattis, **Isabelle S. Goldstein**, and Savvas M. Koushiappas
Phys. Rev. D 102, 061301(R) (2020)

RESEARCH EXPERIENCE

Texas A&M University

Postdoctoral Researcher with Dr. Louis Strigari

2023–Present

- Research focus on dark matter in dwarf galaxies using stellar kinematics.
- Worked with data from the Gaia space observatory.

Brown University

Graduate Student Research Assistant with Dr. Savvas Koushiappas

2018–2023

- Experience in cosmology, astroparticle physics, axion-like dark matter, dwarf galaxy dynamics and primordial black holes.
- Coding fluency in Python, Fortran 90, C++, and Cluster high performance computing.

Carnegie Mellon University

Undergraduate Research Assistant with Dr. Matthew Walker

2017–2018

- Examined the strength of standard dwarf galaxy detection methods using gamma ray data with stellar data from the Sloan Digital Sky Survey and Pan-STARRS.

Lawrence Berkeley National Labs

Assistant Researcher with the DESI and BOSS collaborations

Summer 2016

- Tested ultra faint spectra sky subtraction by integrating the Spectroperfectionism method into DESI and BOSS data analysis pipelines.

Carnegie Mellon University

Undergraduate Student Research Assistant with Dr. Shirley Ho

2015–2016

- Dark matter and galaxy cross correlation bias for redshift dependance.

MENTORING EXPERIENCE

LaVena Tilger, Texas A&M University Undergraduate Student 2025-Present
Mentored in usage of Gaia data, analysis methods, and writing a paper.

Angelica Whisnant, Ohio State University Graduate Student 2025-Present
Mentored in running Jeans analysis code and analyzing results.

Kaitlin Webber, Texas A&M University Graduate Student 2023-Present
Mentored while writing a paper.

Levi K. C. Fisher, University of Hawai'i Undergraduate Student 2024-2025
Mentored in running Jeans analysis code, analyzing results, and publishing a paper.

TEACHING EXPERIENCE

Center for the Integration of Research, Teaching & Learning at Texas A&M

- An Introduction to Evidence-Based Undergraduate Teaching Certificate Fall 2024

Brown University

- Lab Instructor, PHYS 0470 Electricity and Magnetism Fall 2018, 2019, 2020
— Course Instructor: Dr. Savvas Koushiappas
— Number of students: 22, 24, 32
- Lab Instructor, PHYS 0060 & 0160 Spring 2019
— Course Instructor: Dr. Meenakshi Narain
— Number of students: 15
- Lab Instructor, PHYS 0220 Astronomy Spring 2020
— Course Instructor: Dr. Jonathan Pober
— Number of students: 200
- Course Teaching Assistant, PHYS 0070 Analytical Mechanics Spring 2021
— Course Instructor: Dr. James Valles
— Number of students: 85

SCHOLARSHIPS AND AWARDS

- Physics Merit Fellowship
Brown University Department of Physics 2022-2023
- Award of Excellence as a Graduate Teaching Assistant
Brown University Department of Physics, PHYS 0070 2021
- RI Space Grant Graduate Fellow with the NASA RI Space Grant Consortium 2021
- National Science Foundation Graduate Research Fellowship Program Honorable Mention 2020
- Associate Member of Sigma Xi Scientific Research Honors Society
Brown University Chapter, Elected to Membership 2020

- Award of Excellence as a Graduate Teaching Assistant
Brown University Department of Physics, PHYS 0470 2018
- Senior Leadership Recognition Award
Carnegie Mellon University Department of Physics 2018
- NASA Pennsylvania Space Grant
Investigating the CMB lensing potential: stacking with galaxies and gamma radiation density 2015
- H. Joseph Gerber Medal of Excellence
Connecticut Academy of Science and Engineering 2014
- CERN Special Award at the Intel International Science and Engineering Fair
European Organization for Nuclear Research 2014

SEMINARS & COLLOQUIA

- University of Hawai'i at Mānoa, High Energy Physics Seminar
Title: *Constraining dark matter density profiles with dwarf galaxy kinematics: the 2D and the 6D story* 2026
- Black Hills State University, Physics Seminar
Title: *Constraining dark matter density profiles with dwarf galaxy kinematics: the 2D and the 6D story* 2026
- Yale University, Astrophysics Seminar
Title: *Kinematics of the Sagittarius Dwarf Spheroidal core: A 5D Analysis for a 6D Methodology with Gaia DR3* 2025
- University of Oklahoma, High Energy Seminar
Title: *Constraining dark matter density profiles with dwarf galaxy kinematics: the 2D and the 6D story* 2025

CONFERENCES

Conference Presentations or Posters

- iPoster presented at the Roman Symposium: Cosmic Cartography with Roman, STScI 2025
Title: *A Kinematic Analysis of the Sagittarius Dwarf Spheroidal Core: A 5D Analysis for a 6D Methodology*
- Poster presented at Galactic Frontiers II, Dartmouth College 2025
Title: *A Kinematic Analysis of the Sagittarius Dwarf Spheroidal Core: A 5D Analysis for a 6D Methodology*
- Talk presented at the Center for Theoretical Underground Physics and Related Areas 2025
Title: *Kinematics of the Sagittarius Dwarf Spheroidal Core: A 5D Analysis for a 6D Methodology with Gaia DR3*
- Talk presented at the Virgo Consortium Workshop, University of Sussex, UK 2025
Title: *Constraining Dwarf Galaxy Dark Matter Distributions: Spherical Jeans Analyses for 3D Velocity Data*
- Poster presented at Small Galaxies, Cosmic Questions II, Durham University, UK 2024
Title: *Constraining Dwarf Galaxy Dark Matter Distributions: Spherical Jeans Analyses for 3D Velocity Data*
- Talk presented at the Center for Theoretical Underground Physics and Related Areas 2024
Title: *Constraining Dwarf Galaxy Dark Matter Distributions: Spherical Jeans Analyses for Line-of-Sight and 3D Velocity Data*
- iPoster presented at 243rd annual meeting of the American Astronomical Society 2024
Title: *The Viability Of Ultralight Bosonic Dark Matter In Dwarf Galaxies*

Talk presented at the Theoretical Astroparticle and Cosmology Symposium in Texas, Rice University	2023
Title: <i>The Viability Of Ultralight Bosonic Dark Matter In Dwarf Galaxies</i>	
Poster presented at Essential Cosmology for the Next Generation	2016
Title: <i>Cross correlations of the CMB lensing potential and Sloan Digital Sky Survey galaxies</i>	

Conferences Organized

Junior Member of the Scientific Organizing Committee for the Center for Theoretical Underground Physics and Related Areas (CETUP*)	2026
The Institute for Underground Science at SURF	
LOC Co-chair for the Mitchell Conference on Collider, Dark Matter, and Neutrino Physics	2025, 2026
Texas A&M University	
LOC Co-chair for Texas A&M Astrosymposium	2024, 2025
Texas A&M University	
LOC graduate student organizer for the Conference for Undergraduate Women in Physics	2022
Brown University, <i>Cancelled due to Covid-19</i>	

Conferences Attended

LSST Dark Matter Workshop, Kavli Institute for Cosmological Physics at the University of Chicago	2019
Summer School on Cosmology, International Centre for Theoretical Physics	2016

ADDITIONAL EXPERIENCE

Community outreach

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| • Texas A&M Postdoc-Graduate Mentorship Program Mentor | 2023-Present |
| • Speaker for <i>Astronaut Day</i> , a High School student outreach program at Texas A&M University | 2025 |
| <i>Neighborhood mysteries: lighting up the dark</i> | |
| • Guest speaker at the Urban Assembly School for Emergency Management (NYC, NY) | 2022 |
| <i>Pursuing STEM in undergraduate and graduate level education</i> | |
| • Pittsburgh Glass Center teaching assistant for community glassblowing classes and demonstrations | 2015–2018 |